

RingAlert

vibeFORWARD: M-2 | NYC Tech Week | June 7, 2026
Track 02 — Fraud Watch | Step 0: Product Brief

RingAlert maps bank transactions into a knowledge graph and surfaces coordinated fraud rings that slip under threshold-based rules — giving analysts a 3-minute visual case brief with a Bayesian risk score.

Product Name	End User	Track	Success Condition
RingAlert	Fraud Analyst (3 min/case)	Track 02 Fraud Watch	All 12 fraud-ring accounts identified & explained

The Crisis

A bank's fraud ring stayed under every alert threshold — small transactions, circular flows, coordinated accounts. Traditional rule-based systems flag individual anomalies but cannot see the **network**: twelve accounts moving money in deliberate circles, each transaction individually harmless, collectively a fraud ring draining the system undetected.

The fraud analyst has **three minutes per case**. There is no time for SQL queries, graph traversals, or manual cross-referencing. The product must do all of that automatically and surface one clear, explainable answer.

Multi-Agent Architecture

Agent 1 **Datalngestor**

Step 1 — Find It

Reads the raw 5,000-transaction CSV. Normalises fields, resolves unit conflicts, and stores every account entity and transaction edge into Cognee via `cognee.remember()`. Nothing downstream runs until this succeeds.

Tools: Cognee (write)

Agent 2

GraphMapper

Step 2 — Rank It

Retrieves all entities from Cognee via `cognee.recall()`. Builds a directed relationship graph: who sent to whom, total flow per pair, transaction frequency, timing clusters. Calculates a suspicion score for each account pair and stores the enriched graph back to Cognee.

Tools: Cognee (read + write)

Agent 3

PatternDetector

Step 3 — Act On It

Reads the enriched graph from Cognee. Detects three ring patterns: (a) circular flows ($A \rightarrow B \rightarrow C \rightarrow A$), (b) micro-transactions repeatedly just under alert thresholds, (c) coordinated timing across unrelated accounts. Flags each ring with full evidence trail and escalation recommendation. Stores results to Cognee.

Tools: Cognee (read + write)

Agent 4

RiskScorer

Step 4 — Explain It

Reads flagged rings from Cognee. Uses PyMC Bayesian inference to assign a fraud probability with confidence interval (e.g. '94% likely fraud $\pm 3\%$ '). Generates a **downloadable** narrative PDF: what happened, which accounts, why it's fraud, confidence level. Stores scores to Cognee.

Tools: Cognee (read + write), PyMC

Agent 5

CaseBriefWriter

Step 5 — Show It

Reads everything from Cognee for each flagged case. Writes a plain-English 3-line brief: what happened, why it's suspicious, recommended action. Output surfaces in the UI right panel. A judge with zero training can read it and make a decision in under 3 minutes.

Tools: Cognee (read), Claude API

Judging Criteria — How RingAlert Scores 5/5 on Each

#	Criterion	How RingAlert earns it
1	Agents that work	Agent 1 (DataIngestor) runs directly on the provided 5,000-transaction CSV. If the data loads and Cogn
2	Real collaboration via Cognee	Every agent reads from Cognee before acting and writes back after. <code>cognee.remember()</code> on write, cogn
3	Matches your brief	This brief is written conservatively. The product delivers exactly what is described here: 4+ agents, Cogn
4	End user can use it cold	One screen. Upload CSV, press Run. Agent status shows live. Left panel: network graph with suspicious
5	Explainable	Every risk score shows its Bayesian reasoning (e.g. '94% $\pm 3\%$ — 3 circular flows detected, 47 micro-tra

Sponsor & Mandatory Tool Usage

Tool	Role in RingAlert	Status
Cognee	Memory layer — ALL inter-agent handoffs go through cognee.remember() / cognee.recall(). Graph-native storage mirrors	MANDATORY
PyMC	Bayesian fraud probability with confidence intervals on every flagged ring. Outputs 19440 ± 30 in \$price	MANDATORY
Trupeer	5-minute demo video: CSV upload → agents run → fraud ring appears → analyst reads brief → flags for review. Auto-zo	MANDATORY
Geodo	Domain Expert uses Geodo to research real-world entities in the dataset (merchant names, account patterns) to enrich c	MANDATORY
Lovable	Builds the front-end UI: agent status bar, network graph panel, case brief panel, Flag/Dismiss buttons. No-code, deplora	Recommended
LingCode	Browser-based IDE with Claude Code + Codex + Gemini CLI. Used to write, debug, and communicate with agent pipeline.	Recommended

Dataset

Source: kaggle.com/datasets/quantologist/track02-vibeforward-m-agents
Contents: 5,000 bank transactions containing a hidden 12-account fraud ring.
Target: Surface all 12 accounts — judges hold the answer key.

Devpost Submission Checklist

- [] Product Brief PDF (this document)
- [] GitHub repo link (public, with README)
- [] Trupeer video URL (5 minutes, narrated walkthrough)
- [] Track selection: Track 02 – Fraud Watch
- [] Written product description on Devpost